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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

ACH101T – ALLIED CHEMISTRY

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. The necessary condition for optical activity is
   a) Presence of carbon atom
   b) Presence of asymmetric carbon atom
   c) Presence of water molecule
   d) Compound should be polar

2. Oxidation reaction involves
   a) Loss of electron b) Gain of electron
   c) Loss of oxygen atom d) Gain of hydrogen atom

3. The metal atom present in carbonic anhydrase
   a) Zn b) Co c) Fe d) Mg

4. The ligand H_2O is named as
   a) Aqua b) Aquo c) Water d) Ammine

5. Δ H value of an exothermic reaction is always
   a) Negative b) Positive c) Zero d) None of these

6. pH value of a weak acid is around
   a) 1 b) 5 c) 8 d) 14

7. Paracetamol is a/an
   a) Tranquilizer b) Sedative

c) Anaesthetic d) Antipyretic

8. Chloroform is used as
   a) Local anaesthetic b) General anaesthetic
   c) Antipyretic d) Antiseptic

9. An example for addition polymer is
   a) Nylon b) Terylene c) PVC d) Bakelite

10. The monomer unit of the polythene is
    a) $CH_2 == CH_2$ b) $CH_2 == CHCl$
    c) $CH_2 == C(CH_3)CH=CH_2$ d) - $CONH_2$

II. Fill up the blanks

11. Mixture of 1:1+ tartaric acid and ε- tartaric acid are called as _____.

12. According to Werner's theory, primary valencies are satisfied by _____.

13. Henderson equation for acidic buffer is _____.

14. Chemical name of aspirin is _____.

15. An example for energy yielding nutrient is _____.

III. Match the following

16. Secondary structure of protein	a. Vitamin B_{12}
17. Co	b. Antibacterial
18. Buffer	c. α – helix
19. Sulpha drugs	d. Poly isoprene
20. Rubber	e. NH_4Cl and NH_4OH

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Define isoelectric point.

22. Define resolution of optical isomers.

23. Define central metal ion.

24. What is the function of carboxy peptidase?

25. Define heat of neutralization.

26. What are buffers?

27. Give the mode of action of sulpha drugs.

28. What are tranquilizers?

29. Define polymers with examples.

30. Define calorific value of food. Give its unit.

31. What are the types of chemical bond?

32. What are exothermic and endothermic reactions?

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. i) Discuss the primary structure of protein. **(3)**
    ii) Discuss the optical activity of lactic acid. **(3)**
    iii) What are Zwitter ions? **(1)**

34. i) Write the postulates of werner's theory. **(5)**
ii) Draw the structure of chlorophyll. **(2)**

35. i) Discuss the effect of temperature pH on the activity of enzymes. **(2)**
ii) Derive Henderson equation. **(4)**
iii) Define pH. **(1)**

36. i) What are anaesthetics? How are they classified? **(2)**
ii) Give the structure and uses of sulphadiazine and sulphafurazole . **(2+2)**
iii) What are sedatives? Give one example. **(1)**

37. i) Discuss the classification of polymers. **(3)**
ii) What are sustainable biofuels? **(2)**
iii) What are dendrimers? **(2)**

38. i) Discuss the classification of ligands. **(3)**
ii) Define heat of formation. **(2)**
iii) What are oxidation and reduction reactions? **(2)**

39. i) Explain any one method of resolution. **(3)**
ii) Define racemisation. **(1)**
iii) Derive the expression for general acid base catalysis. **(3)**

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